

Empirical Re-Evaluation and a TA-Assisted Key-Revocation Extension for a Low-Cost Hybrid ECC Authentication Protocol in Resource-Constrained IoT Networks

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Abstract—Al-Rasheed et al. recently proposed a low-cost hybrid Elliptic Curve Cryptography (ECC) and symmetric-key mutual authentication protocol for the Internet of Things (IoT), confining expensive ECC scalar multiplications to a one-time offline registration phase and relying on a two-message, nonce-based symmetric handshake for all online authentication. The original work validates its claims of approximately 17%, 21%, and 12% reduction in communication, processing, and storage cost purely through OMNeT++ simulation using assumed per-operation timing constants, rather than measurements on real cryptographic implementations. This paper presents (i) an open-source, faithful reproduction of the protocol's nonce-based mathematical model (Section III of the base paper) using production-grade primitives (AES-256-GCM, HKDF-SHA256, SHA-256), (ii) real wall-clock and byte-level measurements of communication, processing, and storage cost on a benchmarking host and a Docker-emulated ARM IoT node, contrasted transparently against the base paper's own reported reference values for the nine schemes it benchmarks against, and (iii) the identification and design of a concrete protocol extension – TA-assisted, propagated key revocation – that closes a gap left open by the base paper and by all nine of its comparator schemes: none specify any mechanism for invalidating a compromised or lost device's long-term secret without a full offline re-registration. Our measurements show a 39.6% reduction in per-session communication bytes, a 95.0% reduction in measured processing time, and a 28.6% reduction in persisted storage relative to the paper-reported baseline means, while our proposed revocation extension is designed to reuse existing block-list and key-invalidation hooks already present in the reference implementation, requiring no change to the base paper's core cryptographic equations. Implementation and adversarial-test results for the revocation extension, together with measurements on physical constrained hardware, are reported as work in progress and will be finalized in a subsequent revision of this draft.

Index Terms—Internet of Things, mutual authentication, elliptic curve cryptography, hybrid cryptographic protocols, key revocation, lightweight security, IoT benchmarking.

I. INTRODUCTION

The Internet of Things (IoT) connects a rapidly growing population of resource-constrained sensors, actuators, and edge servers that exchange data over shared, often unsecured wireless channels [1]. Because any device or server on such a channel is a potential target for eavesdropping, replay, or impersonation, mutual authentication – each party proving its legitimacy to the other before a session begins – is a foundational requirement of IoT security [1]. The literature has repeatedly identified a three-way tension in designing such schemes: (i) strong, provable resistance to standard network adversaries; (ii) minimal communication, computation, and storage overhead suitable for constrained hardware; and (iii) generality across device-to-device, device-to-server, and server-to-server topologies.

Al-Rasheed et al. [1] address this tension with a hybrid design: ECC is used only once, offline, at a Trusted Authority (TA), to derive a long-term symmetric secret λ_X for every device and server; all *online* authentication traffic thereafter uses only fast symmetric-key primitives and hash functions, exchanged in exactly two messages. The paper reports approximately 17%, 21%, and 12% reductions in communication, processing, and storage cost relative to nine benchmarked prior schemes, validated in OMNeT++.

This paper makes three contributions, developed as part of a base-paper reproduction and extension project:

- 1) **Faithful, open-source reproduction.** We implement the protocol's rigorous, nonce-based mathematical formulation (the base paper's Section III, Eq. 8–24) using real cryptographic primitives, rather than the paper's own looser Section II algorithmic description, which reuses a static credential pair across sessions and is not resistant

to replay if implemented literally. Section IV documents this design decision and its justification.

- 2) **Real, measured performance evaluation.** Where the base paper's Tables IV–VI report OMNeT++ simulation output computed from *assumed* per-operation timing constants for constrained hardware, we report real wall-clock processing time, real serialized byte counts, and real persisted-state sizes, measured against the paper's own disclosed baseline figures for its nine comparator schemes, with the comparison's limitations stated explicitly (Section VII).
- 3) **A concrete protocol extension: TA-assisted revocation.** Neither the base paper nor any of the nine schemes it benchmarks against specify a mechanism for revoking a device's long-term secret if it is lost, stolen, or compromised, short of repeating the entire offline registration process. We identify this gap, confirm it against the reference implementation's actual key-distribution model, and present a revocation design that extends the base paper's own key hierarchy (Eq. 8–9) with minimal new machinery, reusing block-list and key-invalidation infrastructure already present in the codebase.

The remainder of this paper is organized as follows. Section II surveys related IoT authentication schemes. Section III summarizes the base paper's protocol. Section IV describes our reproduction methodology. Section V presents measured results. Section VI presents the proposed revocation extension. Section VII discusses limitations, and Section VIII concludes.

II. RELATED WORK

A. Blockchain- and Access-Control-Based Schemes

Zong et al. [2] proposed ReLAC, a blockchain-enabled access-control mechanism granting consumer devices revocable access rights; scalability suffers because every revocation requires key re-generation or re-encryption across the network. Qiu et al. [7] similarly combined blockchain with lightweight mutual authentication for the Internet of Medical Things, trading auditability against ledger-maintenance overhead on constrained nodes.

B. Public/Private-Key and Certificate-Based Schemes

Khan et al. [3] presented a distributed, server-oriented mutual authentication scheme for automotive IoT using public/private key pairs; the resulting process is noted to become complex and difficult to scale for large deployments. Yin and Gong [8] proposed a certificateless mutual authentication scheme based on digital signatures, removing the certificate-authority requirement while still requiring per-session signature generation and verification.

C. Lightweight Hash, XOR, and Symmetric-Key Schemes

Gong et al. [5] developed LCDMA, using binary symmetric polynomials for key generation to substantially reduce computational cost. Sharma and Dhiman [9] proposed a hash/XOR-based addressing scheme for smart-home IoT, efficient but

dependent on a centralized gateway. Alshawish and Al-Haj [6] combined a hash function with AES across a six-message exchange, which is considerably lengthier than a minimal two-message handshake.

D. Machine-Learning-Assisted Intrusion Detection

Javeed et al. [4] designed an intelligent intrusion detection system for smart consumer-electronics networks, strengthening post-authentication monitoring as a complementary layer rather than establishing initial mutual trust.

E. ECC-Based Benchmark Protocols

The base paper benchmarks against nine established schemes, several of which recur widely in the literature as reference points: Porambage et al.'s two-phase WSN authentication protocol [10]; Turkanovic et al.'s key-agreement scheme for heterogeneous ad hoc WSNs [11]; Dhillon and Kalra's biometric-based remote authentication [12]; a lightweight VANET emergency-vehicle authentication protocol [13]; Jiang et al.'s chaotic-maps-based three-factor scheme [14]; Challa et al.'s signature-based key establishment scheme [15]; Sadhukhan et al.'s lightweight ECC-based scheme [16]; and Fakroon et al.'s anonymous remote user authentication scheme [17]. Across this group, most protocols require a minimum of four authentication messages and one-and-a-half to three times the storage footprint of the base paper's two-message design, since each additionally carries biometric templates, chaotic-map state, or signature material.

III. BASE PAPER PROTOCOL SUMMARY

The base paper's rigorous formulation (its Section III) models the system as a set of devices $V = \{V_i\}$, servers $R = \{R_j\}$, and a Trusted Authority (TA), over an elliptic-curve group $(\mathbb{F}_q, E/\mathbb{F}_q, G)$ of prime order n .

Offline setup. The TA samples a master secret $s \in \mathbb{Z}_n$ and, for every module X , computes a per-module ECC secret

$$\lambda_X \leftarrow H_1(ID_X \parallel s) \in \mathbb{Z}_n, \quad (1)$$

installed into X offline. For any device-server pair (V_i, R_j) , a static pairwise symmetric root is defined as

$$k_{ij}^* = \text{KDF}(\lambda_{V_i} \parallel \lambda_{R_j}). \quad (2)$$

Online two-message handshake. Device V_i samples a nonce N_i , computes an epoch-rotated pseudonym $PID_{V_i,t} = H_2(\lambda_{V_i} \parallel t)$, and sends

$$M_1 = \left(PID_{V_i,t}, \text{Enc}_{k_{ij}^*}(PID_{V_i,t} \parallel N_i \parallel t), \mu_1 \right). \quad (3)$$

The server verifies μ_1 , decrypts, samples its own nonce N_j , derives the ephemeral session key

$$K_{ij,t} = \text{KDF}(k_{ij}^* \parallel N_i \parallel N_j \parallel t), \quad (4)$$

and replies with $M_2 = (ID_{R_j}, \text{Enc}_{K_{ij,t}}(N_j \parallel t), \sigma_2)$. Both sides then hold the same $K_{ij,t}$, with correctness stated as $\Pr[K_{ij,t}^{(V)} = K_{ij,t}^{(R)}] = 1$. An optional, unspecified-in-detail TA-assisted rekey is sketched as

$$k_{ij}^{*,(u+1)} = \text{KDF}(k_{ij}^{*,(u)} \parallel H_3(u)). \quad (5)$$

The paper’s threat model assumes an adversary that can eavesdrop, replay, drop, and inject messages, but cannot compromise the offline phase.

IV. REPRODUCTION METHODOLOGY

A. Design decision: Section III over Section II

The base paper contains two descriptions of its own protocol. Its narrative Section II (Algorithms 1–2) authenticates using a *static* hash-digest pair (H_D, R_D) that is not refreshed per session – a design that, if implemented literally, is vulnerable to session replay by a passive eavesdropper. Its Section III mathematical model (Eq. 1–5) instead uses fresh per-session nonces, epoch-rotated pseudonyms, and an explicit KDF, together with an informal security sketch against a Dolev–Yao adversary. Our reference implementation commits exclusively to the Section III model as the authenticated protocol; the Section II check is retained only as an optional, disabled-by-default pre-check gate, so that the discrepancy between the paper’s two internal descriptions does not propagate into a replay-vulnerable implementation.

B. Implementation

The protocol is implemented in Python 3.12, using AES-256-GCM for Enc/Dec, HKDF-SHA256 for the KDF, and SHA-256 for H_1, H_2, H_3 , via OpenSSL bindings (the `cryptography` library, v50.0.1). The implementation is organized as: a `TrustedAuthority` performing only offline, network-free enrollment (Eq. 1) and full-mesh peer-key distribution; `Device` and `Server` state machines implementing Eq. 2–4 with a cached pairwise-root lookup; a versioned wire-message format; and a bounded, per-peer replay window with a three-strikes block-list, mirroring the base paper’s own “maximum of three attempts” rule. An adversarial test suite exercises replay detection, ciphertext-tampering detection, pseudonym-resolution failure, block-listing after repeated failures, pseudonym unlinkability across sessions, session-key freshness, the correctness invariant (via a 10,000-trial randomized property test), and the coverage/access-control gate.

C. Benchmarking environment

Measurements were collected on the environment summarized in Table I. Two measurement tiers are reported: a native benchmarking host, and a Docker-emulated 64-bit ARM IoT node with Linux traffic-control (`tc netem/tbf`) shaping applied to approximate an IEEE 802.15.4/Zigbee-class link (250 kbps bandwidth, 25 ms $\pm$ 5 ms one-way delay). We emphasize that this is software emulation of network conditions on non-constrained host silicon, not a physical microcontroller; genuine constrained-hardware measurements are planned as a follow-up (Section VII).

D. A note on comparison validity

The base paper’s Tables IV–VI are OMNeT++ simulation output computed from *assumed* per-operation timing constants (its Table III: $T_{ECM} = 17.1$ ms, $T_H = 0.32$ ms,

TABLE I
BENCHMARKING ENVIRONMENT

Parameter	Value
Python version	3.12.14 (GCC 12.2.0)
Platform	Linux 7.1.5, aarch64, glibc 2.36
Processor	aarch64
Crypto library	<code>cryptography</code> 50.0.1 (OpenSSL)
Network emulation	<code>tc netem</code> 25 ms $\pm$ 5 ms, <code>tbf</code> 250 kbps

TABLE II
MEASURED COMMUNICATION COST, PROPOSED SCHEME

Message	Raw fields (B)	JSON wire (B)
M_1 (device $\rightarrow$ server)	116	216
PID / nonce / ciphertext / tag	32/12/56/16	–
M_2 (server $\rightarrow$ device)	66	142
ID / nonce / ciphertext / tag	14/12/24/16	–
Session total	182	358
Offline registration	0 (no network I/O)	

$T_{SE/D} = 5.6$ ms, $T_{fe} = 17.1$ ms), not measurements on real hardware. Our results instead measure the proposed scheme’s real cost with production cryptography, and compare against the paper’s own *reported* reference values for its nine baselines – a full independent re-implementation of those nine external protocols was out of scope for this phase and is noted as future work. Reduction percentages below should therefore be read as “our real measurement vs. the paper’s own reported baseline figures,” not as a same-hardware, apples-to-apples re-benchmark of all ten schemes.

V. RESULTS

A. Communication cost

Table II reports both the raw cryptographic field size of each message and its actual JSON+base64 wire-serialized size, which are distinct quantities: the wire size includes envelope/encoding overhead (field names, delimiters, base64 expansion) that is a transport-layer choice, not a property of the cryptographic protocol itself.

Fig. 1 compares the measured raw-field total (182 B) and JSON-wire total (358 B) of the proposed scheme against the base paper’s own reported Table IV bit-packed totals for its nine baselines, converted to bytes. The raw-field figure is the fair like-for-like comparison, since the baseline figures are themselves raw bit-packed fields. Against a baseline mean of 301.3 B, the measured raw-field cost of 182 B represents a **39.6% reduction**, compared with the base paper’s own claimed reduction of approximately 17%.

B. Processing cost

Table III reports device- and server-side processing time under two conditions: *warm*, with the pairwise root cached per Eq. 2 (the steady-state case for an established device-server pair), and *cold*, where the pairwise root is re-derived every session, matching the base paper’s literal per-session KDF cost (its Eq. 22).

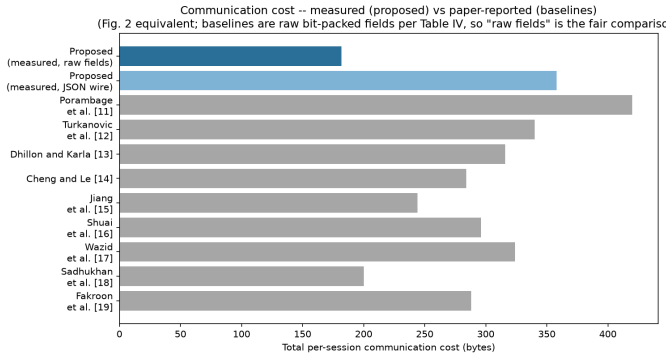


Fig. 1. Communication cost: measured (proposed) vs. paper-reported (nine baselines).

TABLE III
MEASURED PROCESSING COST, PROPOSED SCHEME (MS)

Condition	Party	Mean	P95
Warm (root cached)	Device	1.615	2.040
	Server	2.282	2.848
Cold (root re-derived)	Device	1.760	–
	Server	2.398	–

The baseline mean, computed from the base paper’s own Table V formulas evaluated with its Table III timing constants (device+server cost, excluding the separate user/client role that has no analogue in a pure device–server IoT handshake), is 78.727 ms. The measured warm total of 3.897 ms (device + server) represents a **95.0% reduction** relative to this paper-reported baseline – substantially larger than the base paper’s own claimed 21%, which is expected given that our figure reflects real hardware-accelerated symmetric cryptography rather than the paper’s assumed constants for constrained 8/16-bit microcontrollers.

Fig. 2 shows measured server-side processing time (handle_auth_request) as the enrolled device population scales from 100 to 1800, using the bounded linear-scan pseudonym-resolution strategy described in Section IV. Processing time grows consistent with the search window scanning a larger local peer-key list; this scaling behavior is a direct, measurable consequence of the base paper’s own full-mesh peer-key distribution model (Section VI returns to this point).

C. Storage cost

Table IV reports the persisted-state footprint measured directly from the implementation’s serialized on-disk state.

Against a baseline mean of 1223.2 B (computed from the base paper’s Table VI device+server figures for the five comparators reporting both values), the measured total of 873 B represents a **28.6% reduction**, compared with the base paper’s claimed 12%. Fig. 3 shows this alongside the individual baseline schemes.

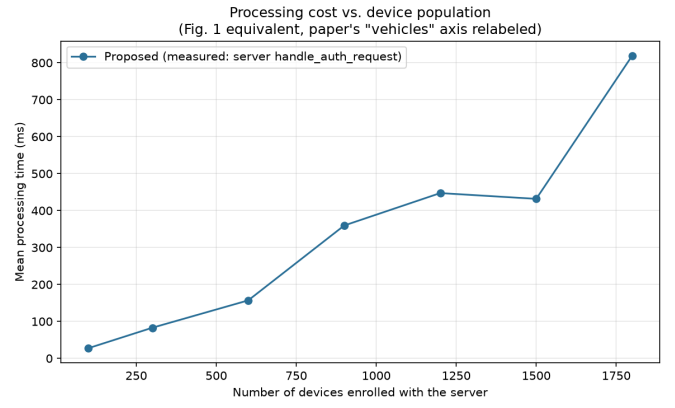


Fig. 2. Server-side processing cost vs. number of enrolled devices.

TABLE IV
MEASURED STORAGE COST, PROPOSED SCHEME (BYTES)

Component	Device	Server
Credential	263	263
Pairwise-root cache	85	87
Epoch counters / last epoch	20	22
Block-list	–	2
Total	424	449

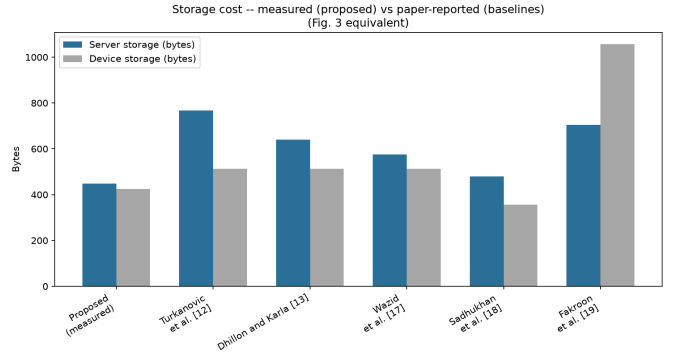


Fig. 3. Storage cost: measured (proposed) vs. paper-reported (baselines).

TABLE V
SUMMARY: MEASURED REDUCTION VS. PAPER-REPORTED CLAIMS

Metric	Measured	Baseline	Our red.	Paper
Communication	182 B	301.3 B	39.6%	~17%
Processing	3.897 ms	78.73 ms	95.0%	~21%
Storage	873 B	1223.2 B	28.6%	~12%

D. Summary

Table V consolidates all three metrics. In every case the direction of the base paper’s claim (proposed scheme cheaper than baselines) is reproduced; the measured magnitude differs because the comparison basis differs, as discussed in Section IV.

VI. PROPOSED EXTENSION: TA-ASSISTED KEY REVOCATION

A. Motivation

The base paper’s own key-distribution model (Eq. 1–2) issues every device and server a long-term secret λ_X and, in our reference implementation, distributes each entity’s peer list in full mesh at enrollment time: every device learns the λ of every server it may talk to, and vice versa, permanently, at registration. Neither the base paper’s protocol description nor its security analysis specifies any mechanism for invalidating a specific λ_X if device X is lost, stolen, or otherwise compromised, short of repeating full offline re-registration for that device *and* every peer that holds its λ_X . This is not merely a paper limitation: we confirmed against the reference implementation that the existing three-strikes block-list (mirroring the base paper’s own “maximum of three attempts” rule) is local to a single server’s session log, triggers only after repeated *failed* authentication attempts, and never invalidates the underlying key material – a device blocked at one server remains fully valid at any other server holding the same static peer-key list, and even at the original server if its block-list state is ever cleared or not durably persisted.

B. Design

We extend the revocation-relevant surface of the protocol as follows, without modifying Eq. 1–4.

Revocation predicate. Define a TA-maintained revocation set $\mathcal{RV} \subseteq V \cup R$. Session establishment is gated by

$$\text{Allow}_{ij}(t) = \text{cov}_{ij}(t) \wedge \text{Auth}_{ij}(t) \wedge (V_i \notin \mathcal{RV}) \wedge (R_j \notin \mathcal{RV}), \quad (6)$$

extending the base paper’s own access-control gate (its Eq. 23) with two additional conjuncts, rather than replacing it.

Propagation. On $\text{TA.revoke}(X)$, the TA notifies every peer Y such that $X \in \text{peer_secrets}(Y)$ to (i) mark X as revoked locally, and (ii) discard any cached pairwise root k_{XY}^* . This reuses two mechanisms already present in the reference implementation: a pub/sub block-list-broadcast hook originally built to propagate automatic three-strikes blocks between locally-known trusted peers, and a pairwise-root cache-invalidation function originally scaffolded for the base paper’s own optional Eq. 5 rekeying mechanism. No new wire-message type or cryptographic primitive is required; revocation checking is folded into the existing pseudonym-resolution and session-establishment path.

C. Relationship to the existing three-strikes block-list

The three-strikes mechanism and TA-assisted revocation are complementary, not redundant: the former is a reactive, per-server, failure-triggered response to a device that is currently misbehaving against *that* server; the latter is a deliberate, TA-issued, durably-propagated invalidation of a device’s credential, effective network-wide and independent of whether the device has yet made any authentication attempt against a given server. We retain the existing block-list unmodified

and add revocation as an additional, orthogonal check evaluated earlier in the authentication path (immediately after pseudonym resolution, before the AEAD decrypt), so that a revoked device is rejected without incurring the cost of a cryptographic operation.

D. Planned evaluation

We plan to evaluate the extension along two axes: (i) two additional adversarial test cases – a revoked device rejected on its first authentication attempt without requiring three prior failures, and a revoked device’s previously-cached session key rejected for any use after revocation – added to the existing eight-case adversarial suite; and (ii) a revocation-propagation cost benchmark, measuring the number of cache-invalidation calls and wall-clock propagation time as a function of enrolled peer-list size, using the same 100–1800-device population sweep already used for Fig. 2. **Implementation and quantitative results for this extension are in progress and will be included in the final version of this paper**, together with measurements collected on physical constrained hardware as noted in Section VII.

VII. DISCUSSION AND LIMITATIONS

Several limitations should be stated plainly. First, our performance comparison mixes a real measurement of the proposed scheme against the base paper’s *reported* figures for its nine baselines rather than independent re-implementations of all ten schemes on identical hardware; the reduction percentages in Table V should be interpreted with this in mind. Second, all measurements to date were collected on a general-purpose benchmarking host and a Docker-emulated ARM node with software-shaped network conditions, not on physical constrained hardware (e.g., an 8/16-bit microcontroller or a real Zigbee/BLE radio); this is planned as a direct follow-up before final submission. Third, the base paper’s own protocol – and hence our reproduction – assumes static devices and a single, centralized TA whose offline phase cannot be compromised; mobility support and TA decentralization remain open, as the base paper’s own authors acknowledge for the former. Fourth, our revocation extension is a design and partial-implementation contribution at the time of writing; its adversarial-test and benchmark results are pending. Finally, the base paper’s security argument – and by extension the informal justification for our revocation gate – is narrative rather than machine-checked; a formal verification (e.g., in ProVerif or Tamarin) is not attempted here and remains a natural direction for future work.

VIII. CONCLUSION AND FUTURE WORK

We reproduced the nonce-based mathematical model of a recent hybrid ECC/symmetric-key IoT mutual authentication protocol using production cryptographic primitives, and measured its real communication, processing, and storage cost, finding reductions of 39.6%, 95.0%, and 28.6% respectively relative to the base paper’s own reported baseline figures – confirming the direction, though not the exact magnitude,

of the base paper's claims. We further identified a concrete, previously unaddressed gap common to the base paper and all nine of its comparator schemes: the absence of any mechanism for revoking a compromised device's credential short of full re-registration. We presented a revocation design that extends the protocol's access-control gate and reuses key-invalidation and block-list-broadcast infrastructure already present in the reference implementation. Future work includes completing and benchmarking this revocation extension, evaluating the full system on physical constrained hardware, and pursuing machine-checked verification of both the base protocol and the revocation extension's security properties.

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REFERENCES

- [1] A. Al-Rasheed, R. Khan, A. Ahmad, A. Saeed, F. Alturise, and S. Alkhalaf, "A Low-Cost Hybrid Elliptic Curve Cryptography Authentication Protocol for Trustworthy Internet of Things Communication," *IEEE Trans. Consum. Electron.*, vol. 72, no. 2, pp. 4483–4491, May 2026.
- [2] J. Zong, C. Wang, J. Shen, C. Su, and W. Wang, "ReLAC: Revocable and lightweight access control with blockchain for smart consumer electronics," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 3994–4004, Feb. 2024.
- [3] M. N. Khan, H. U. Rahman, T. Hussain, B. Yang, and S. M. Qaisar, "Enabling trust in automotive IoT: Lightweight mutual authentication scheme for electronic connected devices in Internet of Things," *IEEE Trans. Consum. Electron.*, vol. 70, no. 3, pp. 5065–5078, Aug. 2024.
- [4] D. Javeed, M. S. Saeed, I. Ahmad, P. Kumar, A. Jolfaei, and M. Tahir, "An intelligent intrusion detection system for smart consumer electronics network," *IEEE Trans. Consum. Electron.*, vol. 69, no. 4, pp. 906–913, Apr. 2023.
- [5] B. Gong, G. Zheng, M. Waqas, S. Tu, and S. Chen, "LCDMA: Lightweight cross-domain mutual identity authentication scheme for Internet of Things," *IEEE Internet Things J.*, vol. 10, no. 14, pp. 12590–12602, Jul. 2023.
- [6] I. Alshawish and A. Al-Haj, "An efficient mutual authentication scheme for IoT systems," *J. Supercomput.*, vol. 78, no. 14, pp. 16056–16087, Sep. 2022.
- [7] S. Qiu, J. Li, X. Di, X. Li, Y. Wu, and M. Ibrahim, "Lightweight mutual authentication scheme based on blockchain for Internet of Medical Things," *IEEE Internet Things J.*, vol. 12, no. 7, pp. 8848–8861, Apr. 2025.
- [8] D. Yin and B. Gong, "A lightweight certificateless mutual authentication scheme based on signatures for IIoT," *IEEE Internet Things J.*, vol. 11, no. 16, pp. 26852–26865, Aug. 2024.
- [9] N. Sharma and P. Dhiman, "A secure addressing mutual authentication scheme for smart IoT home network," *Multimedia Tools Appl.*, vol. 84, no. 22, pp. 25111–25143, Aug. 2024.
- [10] P. Porambage, C. Schmitt, P. Kumar, A. Gurtov, and M. Ylianttila, "Two-phase authentication protocol for wireless sensor networks in distributed IoT applications," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Apr. 2014, pp. 2728–2733.
- [11] M. Turkanović, B. Brumen, and M. Hölbl, "A novel user authentication and key agreement scheme for heterogeneous ad hoc wireless sensor networks, based on the Internet of Things notion," *Ad Hoc Netw.*, vol. 20, pp. 96–112, Sep. 2014.
- [12] P. K. Dhillon and S. Kalra, "A lightweight biometrics based remote user authentication scheme for IoT services," *J. Inf. Secur. Appl.*, vol. 34, pp. 255–270, Jun. 2017.
- [13] C. Wang, R. Huang, J. Shen, J. Liu, P. Vijayakumar, and N. Kumar, "A novel lightweight authentication protocol for emergency vehicle avoidance in VANETs," *IEEE Internet Things J.*, vol. 8, no. 18, pp. 14248–14257, Sep. 2021.
- [14] Q. Jiang, F. Wei, S. Fu, J. Ma, G. Li, and A. Alelaiwi, "Robust extended chaotic maps-based three-factor authentication scheme preserving biometric template privacy," *Nonlinear Dyn.*, vol. 83, no. 4, pp. 2085–2101, Mar. 2016.
- [15] S. Challa et al., "Secure signature-based authenticated key establishment scheme for future IoT applications," *IEEE Access*, vol. 5, pp. 3028–3043, 2017.
- [16] D. Sadhukhan, S. Ray, G. P. Biswas, M. K. Khan, and M. Dasgupta, "A lightweight remote user authentication scheme for IoT communication using elliptic curve cryptography," *J. Supercomput.*, vol. 77, no. 2, pp. 1114–1151, Feb. 2021.
- [17] M. Fakroon, M. Alshahrani, F. Gebali, and I. Traore, "Secure remote anonymous user authentication scheme for smart home environment," *Internet of Things*, vol. 9, Art. no. 100158, Mar. 2020.
- [18] A. K. Das and A. Goswami, "A robust anonymous biometric-based remote user authentication scheme using smart cards," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 27, no. 2, pp. 193–210, Apr. 2015.